# Solution: Problem Challenge 1: Rearrange String K Distance Apart

## Problem Statement

Given a string and a number ‘K’, find if the string can be rearranged such that the same characters are at least ‘K’ distance apart from each other.

**Example 1:**

```
Input: "mmpp", K=2
Output: "mpmp" or "pmpm"
Explanation: All same characters are 2 distance apart.
```

**Example 2:**

```
Input: "Programming", K=3
Output: "rgmPrgmiano" or "gmringmrPoa" or "gmrPagimnor" and a few more
Explanation: All same characters are 3 distance apart.
```

**Example 3:**

```
Input: "aab", K=2
Output: "aba"
Explanation: All same characters are 2 distance apart.
```

**Example 4:**

```
Input: "aappa", K=3
Output: ""
Explanation: We cannot find an arrangement of the string where any two
'a' are 3 distance apart.
```

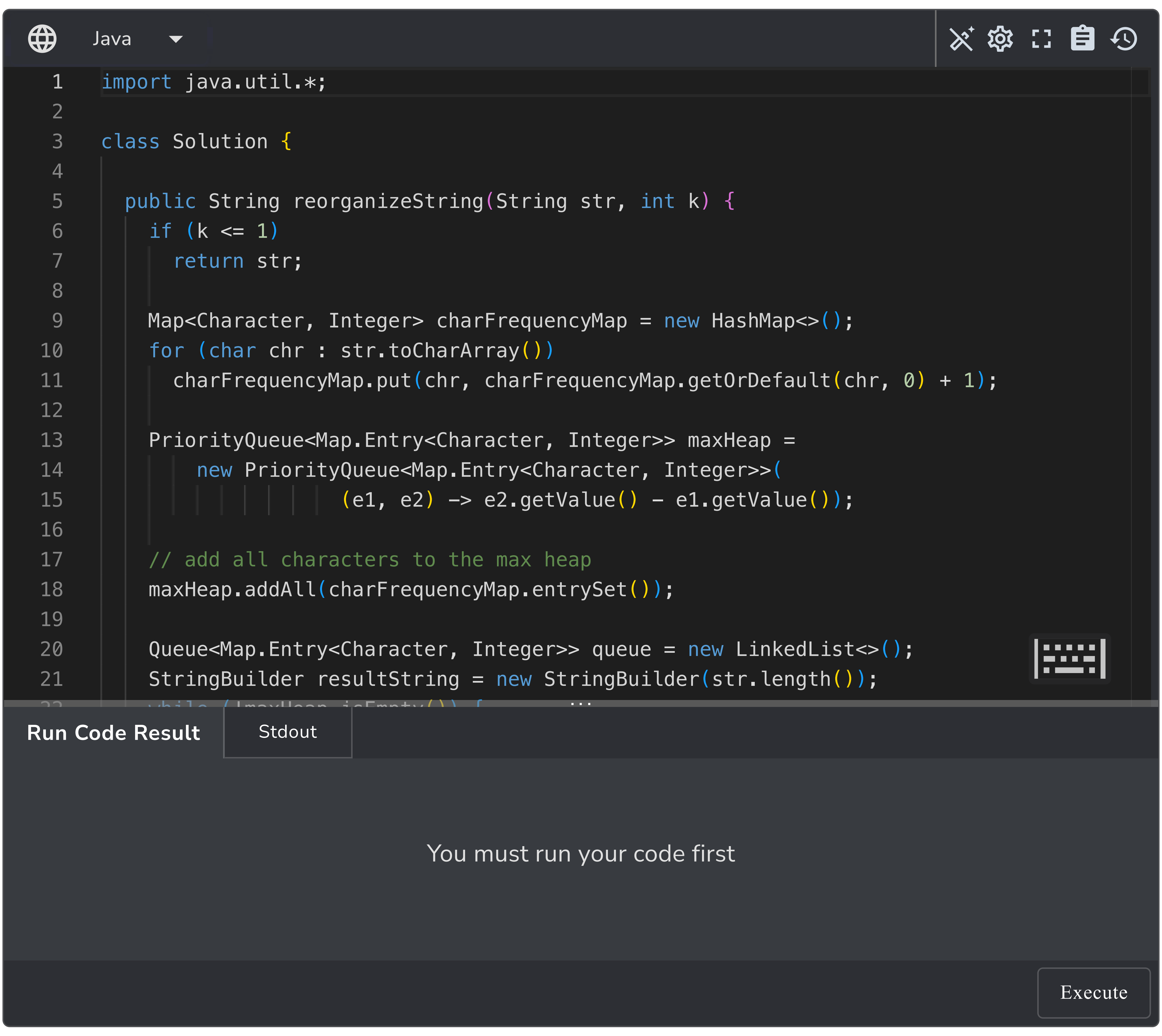
## Solution

This problem follows the `Top ‘K’ Numbers` pattern and is quite similar to `Rearrange String`. The only difference is that in the ‘Rearrange String’ the same characters need not be adjacent i.e., they should be at least ‘2’ distance apart (in other words, there should be at least one character between two same characters), while in the current problem, the same characters should be ‘K’ distance apart.

Following a similar approach, since we were inserting a character back in the heap in the next iteration, in this problem, we will re-insert the character after ‘K’ iterations. We can keep track of previous characters in a queue to insert them back in the heap after ‘K’ iterations.

## Code

Here is what our algorithm will look like:



```java
import java.util.*;

class Solution {

  public String reorganizeString(String str, int k) {
    if (k <= 1)
      return str;

    Map<Character, Integer> charFrequencyMap = new HashMap<>();
    for (char chr : str.toCharArray())
      charFrequencyMap.put(chr, charFrequencyMap.getOrDefault(chr, 0) + 1);

    PriorityQueue<Map.Entry<Character, Integer>> maxHeap =
        new PriorityQueue<Map.Entry<Character, Integer>>(
              (e1, e2) -> e2.getValue() - e1.getValue());

    // add all characters to the max heap
    maxHeap.addAll(charFrequencyMap.entrySet());

    Queue<Map.Entry<Character, Integer>> queue = new LinkedList<>();
    StringBuilder resultString = new StringBuilder(str.length());
```

You must run your code first

## Time Complexity

The time complexity of the above algorithm is O(N∗logN) where ‘N’ is the number of characters in the input string.

## Space Complexity

The space complexity will be O(N), as in the worst case, we need to store all the ‘N’ characters in the HashMap.

← Previous
Problem Challenge 1: Rearrange String K Distance Apart (Hard)

Next →
Problem Challenge 2: Scheduling Tasks (Hard)

☐ Mark as Completed